

FULL NAME: _____

BLOCK: _____

Adding & Subtracting Rational Expressions + Resistance DOK-3 (Q#15 Prep)

Lesson 4-4 · Quick · Teacher Edition

OBJECTIVES

Math Objective	Add and subtract rational expressions with like and unlike denominators; identify the LCM of polynomial denominators; simplify results.
Language Objective	Explain why adding rational expressions requires common denominators, using the word 'LCM' and referring to the analogy with fractions.
Essential Understanding	Rational expressions behave like rational numbers — closed under addition and subtraction, and requiring a common denominator before you can combine them.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Fluency adding/subtracting rational expressions; understanding closure under these operations (Topic 4 assessment Q#15 prep).
Essential Question	When you add two rational expressions with different denominators, why must you find the LCM first?
Materials	Student packet, pencil, calculator. DOK-3 Practice #32 (electrical resistance) is the capstone and rehearses the structural reasoning tested on Topic 4 LEHS Q#15.

[PIN] PRE-FLIGHT — SINGLE-PERIOD COMPRESSED LESSON + DOK-3 CAPSTONE

This is a single-period quick treatment of Lesson 4-4. TWO examples are modeled in Launch (deviation from standard Klimsara one-example rule — adding and subtracting are distinct assessment skills). Release promptly at minute 18.

Today's capstone is Practice #32 (electrical resistance parallel circuit). Students must (a) recognize that $1/R_{\text{total}} = 1/R_A + 1/R_B$ is a sum of rational expressions, (b) build a common denominator, (c) form the ratio of two such resistances as a single simplified rational expression. Same structural reasoning as Topic 4 LEHS Q#15.

Pace: Try-It 3 (4 min) → Try-It 4 (4 min) → #12 (3 min) → #16 (4 min) → #26 (5 min) → #32 (10 min). ~30 min total Explore (2 min buffer for transitions).

45-min Wed F variant: cut #16 + #26. Never cut #32.

Tag: bridge from 4-3 to adding rational expressions

Do Now**DOK: 1****Minutes: 5****Teacher does**

- Hand out at door.
- Circulate 3 min.
- Cold-call 1 student on the $1/2 + 1/3$ conflict resolution.

Students do

- Read conflict and decide which student's method is correct.
- Justify with the word 'common denominator'.

Questions to ask

- What's the LCM of 2 and 3?
- Why can't you add $1/2 + 1/3$ by adding numerators and denominators?

Adult role

Solo teacher: circulate silently. No hints.

[MIC] BRIDGE SCRIPT (read aloud after Do Now)

"Yesterday (Lesson 4-3) you practiced MULTIPLYING and DIVIDING rational expressions. Today you'll ADD and SUBTRACT them — which requires one new move that multiplication didn't: finding a common denominator first. The Model & Discuss prompt shows two students adding $1/2 + 1/3$ in conflicting ways. Which is right?"

Do Now (4-4-savvas-model-discuss-lesson-4-4-launch)

CRITIQUE & EXPLAIN

Teo and Evander find the following exercise in their homework:

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{9}$$

[IMAGE: A tablet displaying the math problem $1/2 + 1/3 + 1/9$]

- A. Teo claims that a common denominator of the sum is $2 + 3 + 9 = 14$. Evander claims that it is $2 \cdot 3 \cdot 9 = 54$. Is either student correct? Explain why or why not.
- B. Find the sum, explaining the method you use.
- C. Construct Arguments Timothy states that the quickest way to find the sum of any two fractions with unlike denominators is to multiply their denominators to find a common denominator, and then rewrite each fraction with that denominator. Do you agree?

[CHECK] ANSWER KEY

Answer key (from source): A. Only Evander is correct. A common denominator will have each other denominator as a factor. B. $\frac{17}{18}$; converted each fraction to a fraction with denominator 18. C. No; It may be faster to use the least common denominator rather than multiplying the denominators together.

[TARGET] SENTENCE FRAME

"To add $1/2 + 1/3$, I first find a common denominator, which is _____. Then $1/2 = \underline{\hspace{1cm}}/6$ and $1/3 = \underline{\hspace{1cm}}/6$, so the sum is _____. The student who got _____ was wrong because _____."

Tag: teacher models Example 3 + Example 4 [COMPRESSED — 2 examples]

Launch

DOK: 1–2

Minutes: 13

Teacher does

- Project Example 3 (add unlike denominators). Think aloud: factor each denominator, identify LCM, rewrite each fraction.
- "The LCM takes every distinct factor at its highest power."
- Transition to Example 4 without stopping — announce: "Subtracting looks identical UNTIL the distribute step. Watch the minus sign."
- Project Example 4. Think aloud: after combining numerators, distribute the negative across EVERY term.
- Flag the Rules card — students copy into margin.
- 30-sec pause: "Turn to partner: where does the minus sign trap catch students? What do you do to avoid it?"
- Do NOT solve Try-Its or Practice items — release at minute 18.

Students do

- Watch Example 3 silently. Note the LCM-building procedure.
- Watch Example 4. Mark the sign-distribution step.
- During pause: explain sign trap to partner using Rule 2(d).
- Copy Rules card into margin.

Questions to ask

- What's the LCM of $(x - 2)$ and $(x + 3)$?
- How do we rewrite $5/(x - 2)$ so it has denominator $(x - 2)(x + 3)$?
- When we subtract, where does the minus sign apply — to the first term only, or every term of the second numerator?
- After combining, what's the LAST thing you check before you finish?

Adult role

Project both examples back-to-back. Release promptly at minute 18.

[BOOK] RULES — ADD/SUBTRACT RATIONAL EXPRESSIONS (keep printed on your page)

1. LIKE denominators: combine numerators, keep the denominator, simplify.
2. UNLIKE denominators:
 - (a) factor each denominator,
 - (b) find the LCM,
 - (c) rewrite each fraction with the LCM as denominator,
 - (d) combine numerators,
 - (e) simplify.
3. The LCM of two polynomial denominators includes every DISTINCT factor, taken to its highest

power across both.

4. Always check whether the final expression can be simplified by cancelling a common factor.

[WARN] COMPRESSED LESSON — TWO EXAMPLES IN LAUNCH

This is a single-period quick treatment. Teacher models BOTH Example 3 (add unlike denominators) AND Example 4 (subtract unlike denominators) during Launch.

Watch carefully on Example 4 — when you subtract, you must distribute the minus sign across every term of the second numerator. This is the #1 error source.

[MIC] TEACHER SCRIPT

1. "Watch me add two rational expressions with different denominators."
2. "Step 1: Factor each denominator. Step 2: Find the LCM."
3. "Step 3: Rewrite each fraction with the LCM as denominator — multiply top and bottom by the missing factors."
4. "Step 4: Combine numerators. Step 5: Simplify."
5. "Now Example 4 — SUBTRACTION. Same setup, but watch the minus sign distribute across EVERY term of the second numerator."
6. "Today's DOK-3 task uses these moves on a parallel-circuit resistance formula. Same algebraic move as Topic 4 LEHS Q#15."
7. Release to Explore.

Tag: TPS + DOK-3 driver

Explore

DOK: 2–3

Minutes: 32

Teacher does

- 5 laps across 32 min.
- Lap 1 (4 min): Try-It 3 — add unlike. Press pairs on LCM construction.
- Lap 2 (4 min): Try-It 4 — subtract unlike. Watch for sign-distribution errors.
- Lap 3 (3 min): #12 — like denominators (quick confidence build).
- Lap 4 (4+5 min): #16 + #26. #26 is the sign-trap item — press "where does the minus go?" on every pair.
- Lap 5 (10 min): [STAR] #32 Electrical Resistance. Students build common denominators for reciprocal sums, then form the ratio.
- Do NOT give #32 CER answers. Pairs present argument at Share.

Students do

- 6 items in order: Try-It 3 → Try-It 4 → #12 → #16 → #26 → #32.
- For #32: BEFORE computing, write out $1/R_{\text{total}} = 1/R_A + 1/R_B$ for each circuit and identify the LCM.
- Justify the ratio form with sentence frame.

Questions to ask

- Try-It 3: what's the LCM of the two denominators?
- Try-It 4: where does the minus sign distribute?
- #12: like denominators — what's the one step you can skip compared to unlike?
- #16: factor both denominators first. Now what's the LCM?
- #26: watch the subtraction. Where do students typically lose the sign?
- #32: what's the LCM for $1/R_1 + 1/R_2$? How do you flip to get R_{total} ?
- #32: how do you form the ratio R_A/R_B as a SINGLE simplified rational expression?

Adult role

Solo teacher: 5 laps. On #32, press the LCM-of-reciprocals move and the final single-expression form — those are the assessment-critical moves.

Try It 3

Find the sum.

- a. $\frac{x+6}{x^2-4} + \frac{2}{x^2-5x+6}$
- b. $\frac{2x}{3x+4} + \frac{4x^2-11x-12}{6x^2+5x-4}$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): a. $\frac{x+7}{(x+2)(x-3)}$, for $x \neq -2, 2$, and 3 . b. $\frac{8x^2-13x-12}{(2x-1)(3x+4)}$, for $x \neq -\frac{4}{3}$ and $\frac{1}{2}$.

Try It 4

Simplify.

- a. $\frac{1}{3x} + \frac{1}{6x} - \frac{1}{x^2}$
- b. $\frac{3x-5}{x^2-25} - \frac{2}{x+5}$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): a. $\frac{x-2}{2x^2}$, for $x \neq 0$. b. $\frac{1}{x-5}$, for $x \neq -5$ and 5 .

Practice #12

Generalize Explain how addition and subtraction of rational expressions is similar to and different from addition and subtraction of rational numbers.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): The same basic procedures are followed for both. First, find a common denominator, then rewrite the expressions using the common denominator, perform the addition and/or subtraction, and then simplify.

Practice #16

Look for Relationships Find the LCM of $3x^2 + 7x + 2$ and $9x + 3$.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): $3(3x + 1)(x + 2)$.

Practice #26

On Saturday morning, Ahmed decided to take a bike ride from one end of the 15-mile bike trail to the other end of the bike trail and back. His average speed the first half of the ride was 2 mph faster than his speed the second half. Find an expression for Ahmed's total travel time. If his average speed for the first half of the ride was 12 mph, how long was Ahmed's bike ride?

[IMAGE: Bike trail with cyclist, sign "Trail length 15 miles", one label "Speed $x+2$ " and another label "Speed x ".]

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $\frac{30(x+1)}{x(x+2)}$; 2.75 hours.

Practice #32 [STAR] DOK-3 ELECTRICAL RESISTANCE

Model With Mathematics The resistance of electrical circuits A and B are shown. Find the ratio of the resistance of Circuit A to Circuit B.

$$\frac{x^2 - 25}{x - 4} \Omega \quad \text{A} \quad \text{---} \left| \right| \text{---} \quad \text{B} \quad \text{---} \boxed{} \text{---} \frac{x + 5}{x^2 - 16} \Omega$$

[STAR] DOK-3 ELECTRICAL RESISTANCE — Practice #32 (Topic 4 LEHS Q#15 prep)

q32 is the electrical resistance problem. "Two circuits, A and B. You're asked for the ratio of their resistances, where each resistance is expressed as a sum of reciprocals (parallel resistors). The mathematical move is: build a common denominator, add the reciprocals, then form the ratio."

CER expectation: (C) ratio as a single simplified rational expression; (E) the algebra steps — common denominator, add fractions, simplify; (R) the resistance-sum formula $1/R_{\text{total}} = 1/R_1 + 1/R_2$ for parallel circuits, with the observation that the ratio's form depends only on the constituent resistances.

If students get stuck: remind them parallel resistance combines via reciprocal sums (same structural move as adding rational expressions). Ask: what's the LCM of the two denominators? Once they have that, the rest is algebra.

Sentence frame required for ELL.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): $((x - 5)(x + 4) : 1)$ or $(x^2 - x - 20 : 1)$.

Tag: academic conversation + Topic 4 LEHS Q#15 bridge

Share / Summary

DOK: 2

Minutes: 5

Teacher does

- Cold-call 1 pair to present their resistance solution using sentence frame.
- Class signals thumbs. Write the LCM step on board.
- Topic 4 bridge: "This is the structural reasoning on Topic 4 LEHS Q#15. You have now rehearsed it."

Students do

- Presenting pair reads their solution and cites the LCM step.
- Rest of class signals and writes takeaway in margin.

Questions to ask

- Summarize: how did you get from $1/R_1 + 1/R_2$ to a single rational expression?
- Why does the ratio simplify the way it does?

Adult role

Cold-call a pair that struggled on #32 — hold them accountable to the LCM step.

Sentence frames:

- Pair share (Resistance): "For the parallel circuit, $1/R_{\text{total}} = 1/R_1 + 1/R_2$. I found a common denominator of ____, added the reciprocals to get $1/R_{\text{total}} = ____$, then flipped to get $R_{\text{total}} = ____$. The ratio R_A/R_B simplified to ____ because ____."
- Whole class: one pair reads their resistance solution. Class signals thumbs up/sideways.

[SEARCH] BRIDGE TO TOPIC 4 LEHS Q#15

The resistance move — find LCM, combine reciprocals into a single rational expression — is the structural reasoning tested on Topic 4 LEHS Q#15 (operations/closure). You have now rehearsed it.

Tag: summary recap (not graded)

Exit Ticket

DOK: 1–2

Minutes: 3

Teacher does

- Collect at door.
- Scan stem 2 (reciprocal-sum connection). Plan re-teach if many students can't explain the parallel to adding rational expressions.

Students do

- 3 sentence stems, honest.

Questions to ask

- Did students connect LCM to the addition procedure?
- Did students recognize the structural parallel between parallel circuits and rational-expression addition?

Adult role

Collect. No grade.

[CLIPBOARD] EXIT TICKET — Student Form

To add rational expressions with unlike denominators, I must first find the _____ of the denominators.

Practice #32 (resistance) showed me that reciprocal-sum problems (like $1/R_1 + 1/R_2$) use the same algebraic steps as adding rational expressions because _____.

One biggest thing I learned today: _____

[CLOCK] PERIOD A / PERIOD F — 65-MIN CLASS NOTES

Both sections should have 65 min for this lesson. Use the extra 10 min on Explore #32 (Resistance) — the richer the LCM argument, the better the Topic 4 LEHS Q#15 payoff.

45-min Wed F variant: cut #16 + #26 from Explore. Never cut #32.

If finished early: hand out Practice #34 (compound-fraction multi-select) as fast-finisher.

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Add/Subtract Rational Expressions Rules card as a sticker/cut-out.
- For #32, provide the setup: " $1/R_{\text{total}} = 1/R_1 + 1/R_2$ for each circuit. Find a common denominator for each side."
- Accept verbal explanation of the LCM step in lieu of written steps.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames on student page (don't skip).
- Word cards: "LCM" = least common multiple — the smallest expression both denominators divide into; "rational expression" = a fraction of polynomials; "closure" = if you add two of them, you get another one of the same kind.
- "Parallel circuit" = a circuit where the current splits across multiple resistors.
- Pair ELL students with English-dominant partners for TPS.